

MIT ACADEMY OF ENGINEERING , ALANDI – D PUNE

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☐ **20 Problem statements for WordNet dataset using Numpy and Pandas .**

```
Import pandas as pd
Import numpy as np

df=pd.read_csv("wordnet.csv")
```

1. COUNT TOTAL NUMBER OF UNIQUE WORDS

```
print(df['word'].nunique())
```

2. FIND NUMBER OF WORDS IN EACH PART OF SPEECH

```
print(df['pos'].value_counts())
```

3. FIND THE MOST FREQUENT PART OF SPEECH

```
print(df['pos'].mode())
```

4. CALCULATE AVERAGE LENGTH OF WORDS USING NUMPY

```
print(np.mean(df['word'].apply(len)))
```

5. FIND LONGEST WORD IN DATASET

```
print(df.loc[df['word'].apply(len).idxmax()])
```

6. COUNT WORDS WITH MORE THAN 10 CHARACTERS

```
print(len(df[df['word'].apply(len)>10]))
```

7. FIND WORDS THAT HAVE MORE THAN 3 SYNONYMS

```
print(df[df['synonyms'].apply(lambda x:len(x.split(',')>3)])
```

8. FIND AVERAGE NUMBER OF SYNONYMS PER WORD

```
print(np.mean(df['synonyms'].apply(lambda x:len(x.split(',')))))
```

9. EXTRACT ALL NOUNS FROM DATASET

```
print(df[df['pos']=='noun'])
```

10. FIND WORDS WITHOUT HYPERNYMS

```
print(df[df['hypernyms'].isnull()])
```

11. COUNT HOW MANY WORDS HAVE HYPERNYMS

```
print(df['hypernyms'].notnull().sum())
```

12. FIND TOP 5 WORDS WITH MAXIMUM HYPONYMS

```
df['hypo_count']=df['hyponyms'].apply(lambda x:len(str(x).split(',')))
```

13. FIND DUPLICATE WORDS

```
print(df[df.duplicated('word')])
```

14. GROUP WORDS BY POS

```
df.groupby('pos')['syn_count'].mean()
```

15. FIND DEFINITIONS CONTAINING ANIMAL

```
print(df[df['definition'].str.contains('animal',case=False)])
```

16. STANDARD DEVIATION OF WORD LENGTHS

```
print(np.std(df['word'].apply(len)))
```

17. CREATE WORD LENGTH COLUMN

```
df['word_length']=df['word'].apply(len)
```

18. CORRELATION BETWEEN WORD LENGTH

```
print(df[['word_length','syn_count']].corr())
```

19. WORDS STARTING WITH A

```
print(df[df['word'].str.startswith('a')])
```

20. DISTRIBUTION OF WORD LENGTHS

```
print(np.histogram(df['word'].apply(len),bins=5))
```